Trajectory Estimation for a Venus Aerobot Balloon during Deep Space Network Occlusion

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Abstract

At Venus, absolute position is normally recovered through radiometric tracking by the Deep Space Network (DSN). When a balloon drifts onto the far side of the planet, Venus occults the line of sight: a blackout of up to 65 h, during which about 14 508 km are flown with no contact. This work estimates the trajectory of a Venus aerobot across the occlusion by fusing an IMU, a Sun sensor, a radar altimeter and a barometer. A Multiplicative Extended Kalman Filter tracks the attitude and a particle filter tracks the position, coupled in a cascade that re-injects every transferred uncertainty as an inflated measurement covariance. Monte Carlo campaigns of 100 runs give a position RMSE of 26.7 km, never worse than 34.3 km, within the 50 km requirement. A single Sun channel raises convergence from about 60% to 98–99%, while the accuracy is set by the onboard atmospheric model rather than by the Sun sensor.

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1. INTRODUCTION AND STATE OF THE ART

1.1 Venus and the Balloon Mission



Figure 1. Venus seen by Mariner 10: a global, unbroken deck of sulfuric-acid clouds hides the surface from visible-wavelength instruments (NASA/JPL-Caltech).

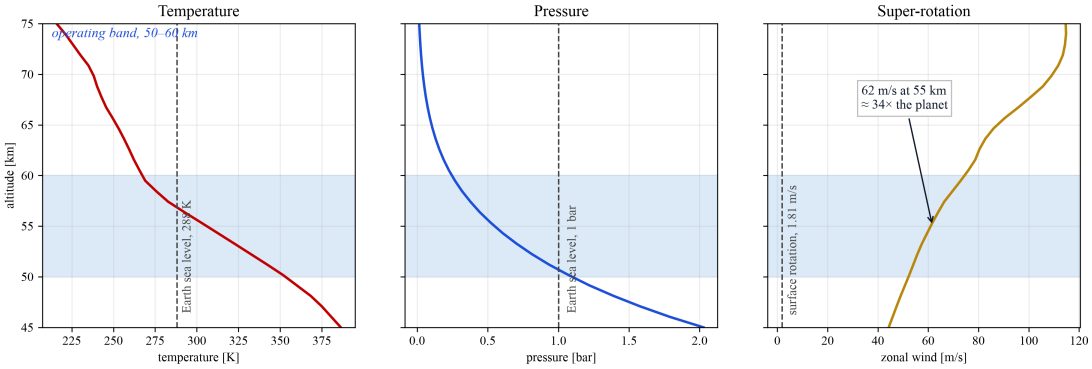


Figure 2. Temperature, pressure and zonal wind between 45 and 75 km (Venus Climate Database). At 50–60 km conditions approach Earth's, with a zonal wind of about 62 m/s.

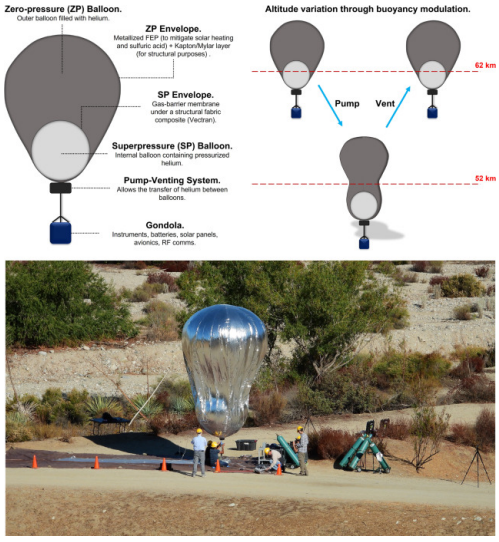


Figure 3. The two-part variable-altitude aerobot: helium pumped into the inner envelope makes the vehicle descend, venting it makes it rise (52–62 km). Adapted from Hall et al.

1.2 The Localization Problem

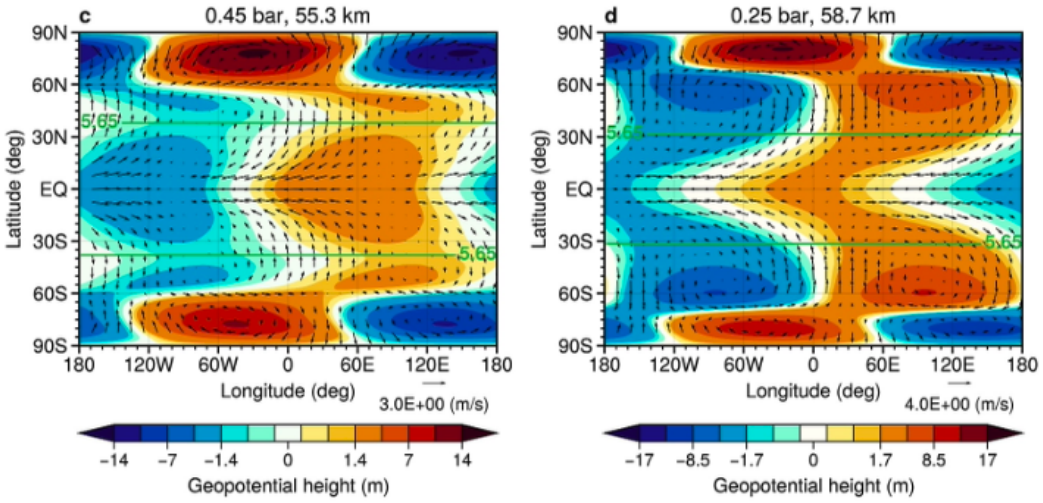


Figure 4. Planetary-scale Rossby–Kelvin wave: wavenumber-one geopotential anomaly sweeping around the planet with a 5.65-day period (Lai et al.).

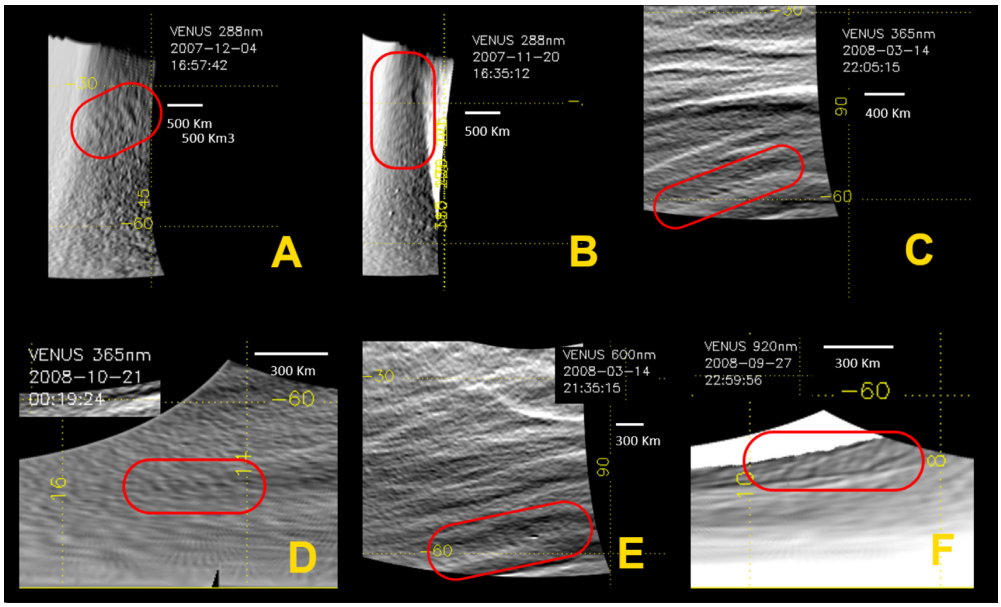


Figure 5. Meso-scale gravity-wave packets in VIRTIS-M cloud imagery, with wavelengths of tens of km and not tied to any topography (Silva et al.).

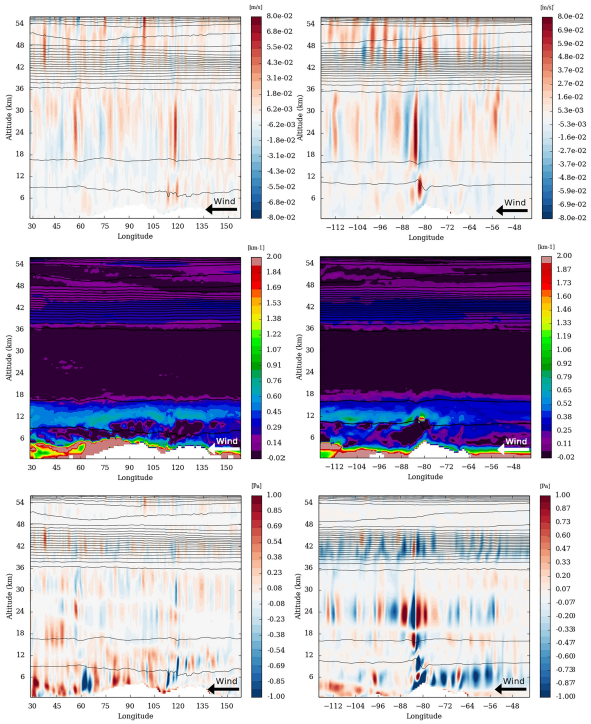


Figure 6. Orographic waves: longitude–altitude cross-sections of vertical wind, Scorer parameter and pressure perturbation, anchored to the topography (Lefèvre et al.).

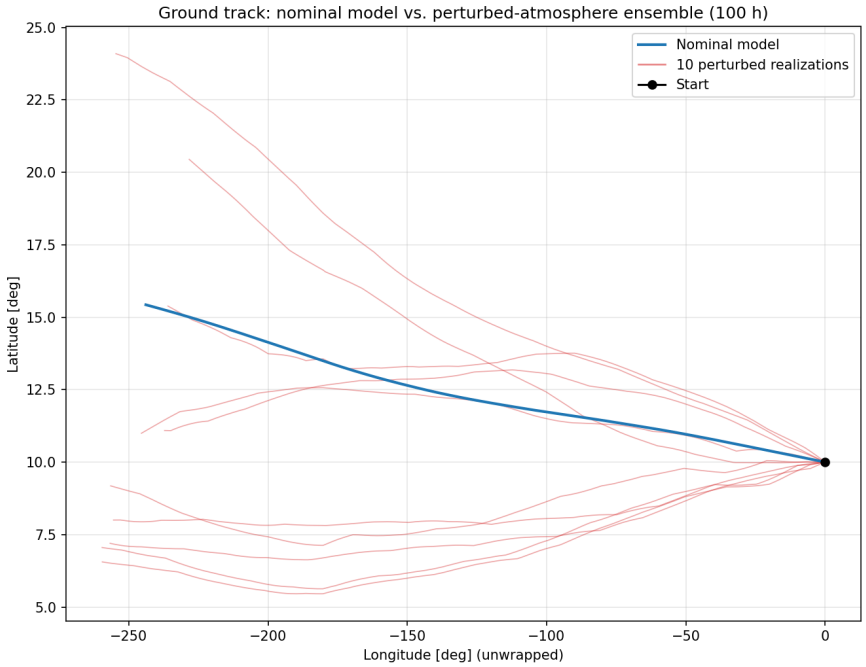


Figure 7. Nominal mean-field ground track vs. ten wind-perturbed realisations from the same initial state: the error grows at about 14 km/h, roughly 900 km at the end of the 65 h blackout.

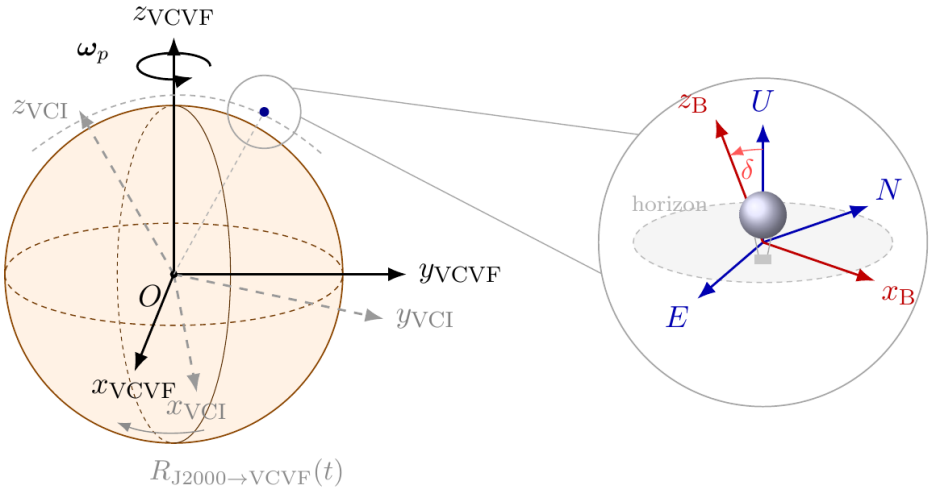
1.3 State of the Art

Table 1. Navigation architectures for a Venus balloon, compared with this work.

	Ploen et al.	Cottrill and Gu	This work
Approach	Bias-compensated inertial navigation, rotated IMU	Inertial navigation, map matching and particle filter, 15 states, 5000 particles	Coupled MEKF and particle filter, 9-state attitude error and 6-state position, 1000 particles
Measurements	IMU only, with orbiter updates	IMU, altimeter, barometer	IMU, altimeter, barometer, Sun sensor
Maps	None	VCD, Magellan topography	VCD, Magellan topography
Horizon	10 min (covariance study)	1 h, 120 Monte Carlo trials	65 h of far-side autonomy, 100 Monte Carlo trials
Reported error	Not reported	47.95 km position RMSE	26.7 km position RMSE

2. DYNAMIC MODEL AND SENSOR SIMULATION

2.1 Reference Frames and Time Systems



(a) planetary frames (b) frames at the vehicle

Figure 8. Reference frames: (a) inertial VCI/J2000 and planet-fixed VCVF; (b) local ENU frame and body frame at the vehicle, tilted by δ from the local vertical.

2.2 Balloon 6-DoF Dynamics

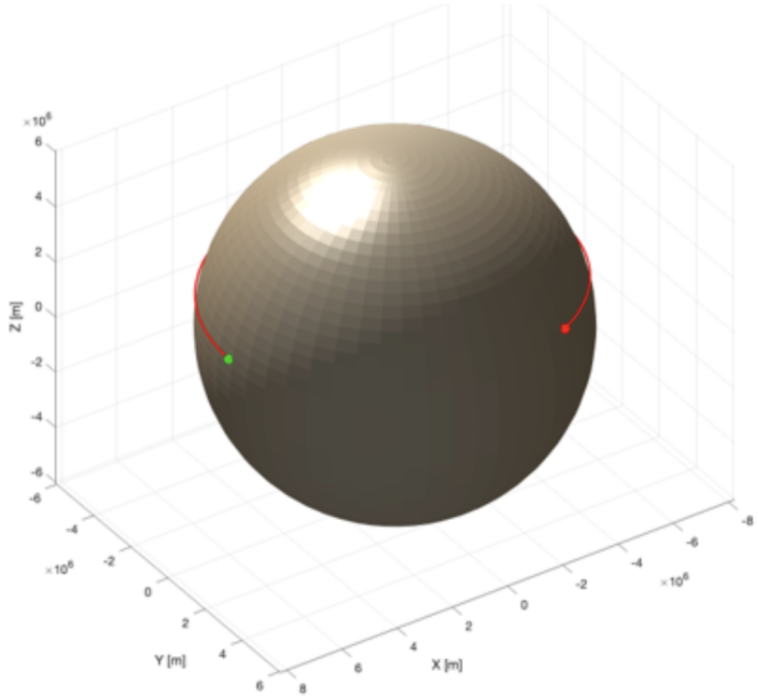


Figure 9. Ground-truth trajectory over the 100 h reference run in VCVF coordinates (green: start, red: end).

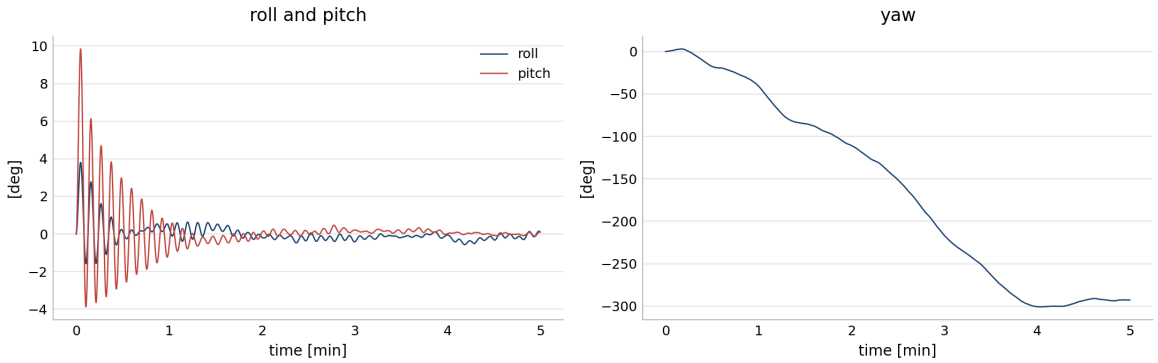


Figure 10. Truth attitude over five minutes. Left: roll and pitch (damped pendulum, within about 1° of the vertical). Right: yaw, which has no restoring moment and drifts freely.

2.3 Simulation Environment

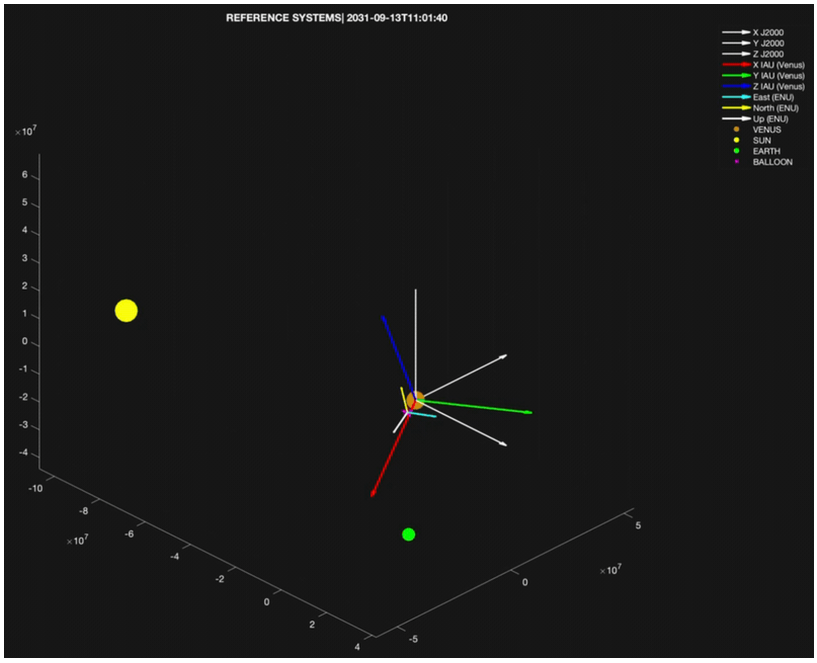


Figure 11. Space environment at the start epoch from SPICE: positions of Sun, Earth and balloon relative to Venus, with J2000, IAU_VENUS and ENU axes.

2.4 Sun Sensor

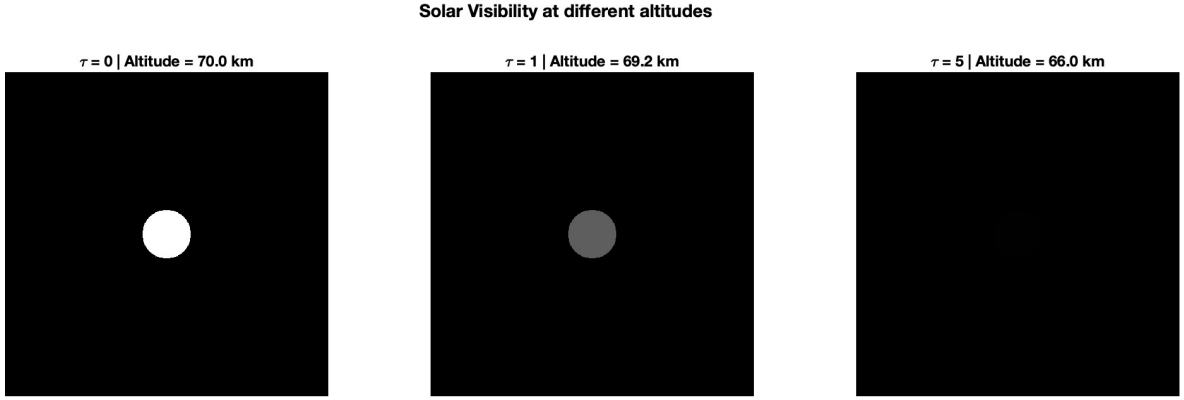


Figure 12. Solar disk seen from below the cloud deck at 70, 69.2 and 66 km: the direct beam is invisible at 66 km, so an optical Sun sensor is not feasible.

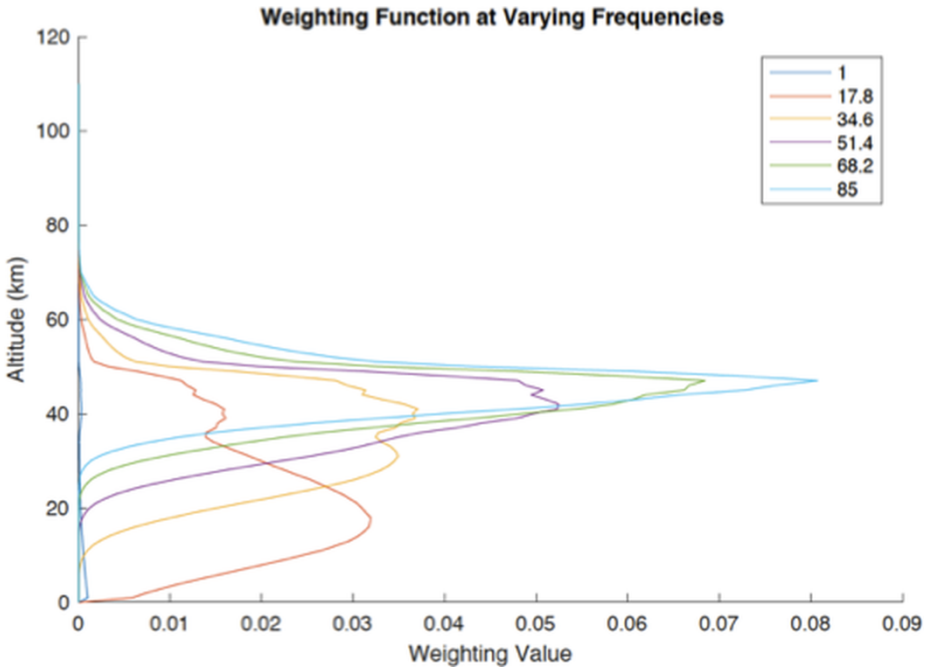


Figure 13. Weighting functions of the Venus atmosphere between 1 and 85 GHz: below about 18 GHz the atmosphere above the balloon is transparent to the Sun, which motivates a microwave Sun sensor.

2.5 Radar Altimeter

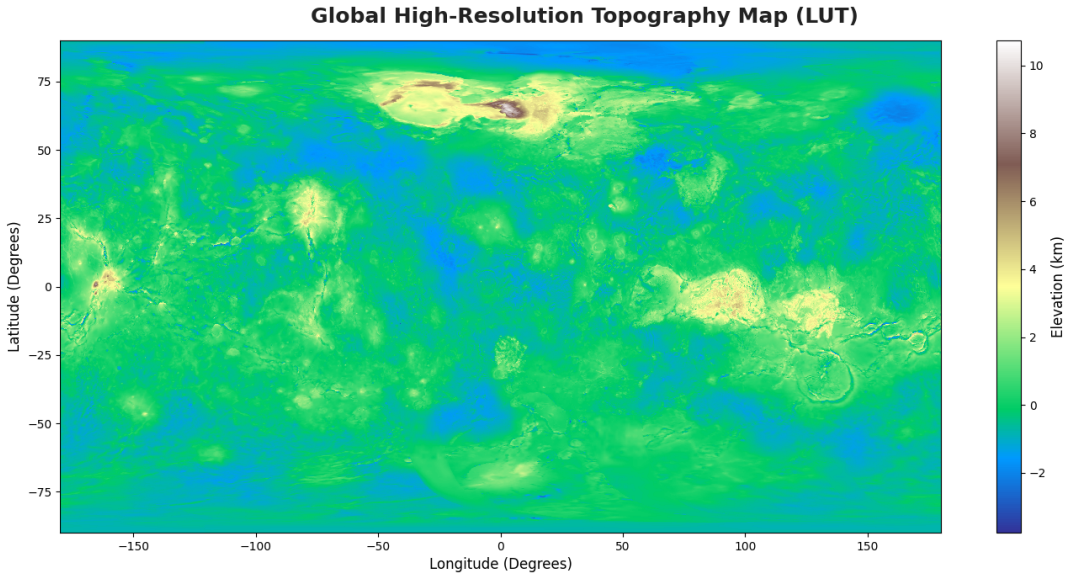


Figure 14. Magellan global topography used as the altimeter look-up table. Aphrodite Terra and Ishtar Terra make the altimeter informative; the plains are where it is weakest.

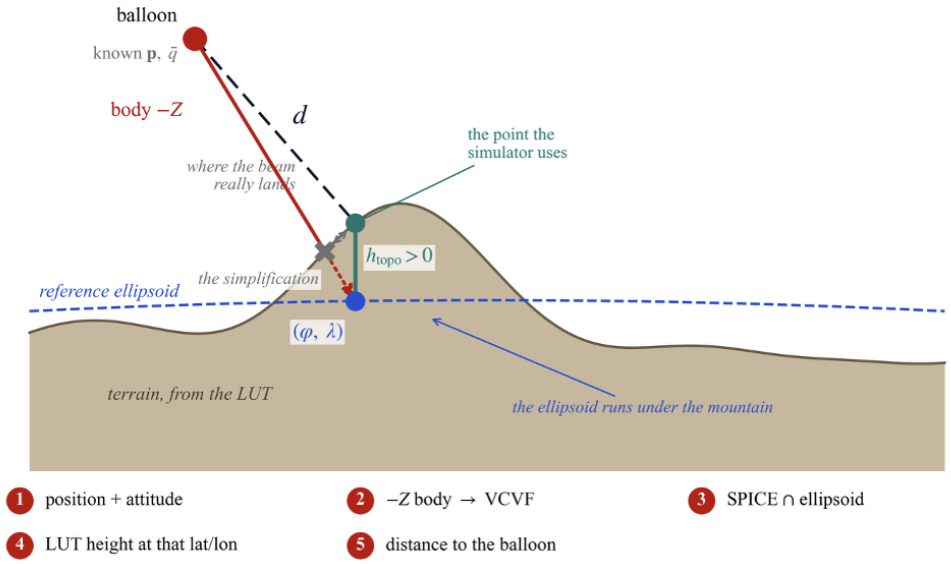


Figure 15. Altimeter ray casting along the body $-Z$ axis against the reference ellipsoid; the terrain elevation at the hit point sets the slant range.

2.6 Summary of the Measurement Set

3. ESTIMATION ARCHITECTURE

3.1 The Cascade Architecture

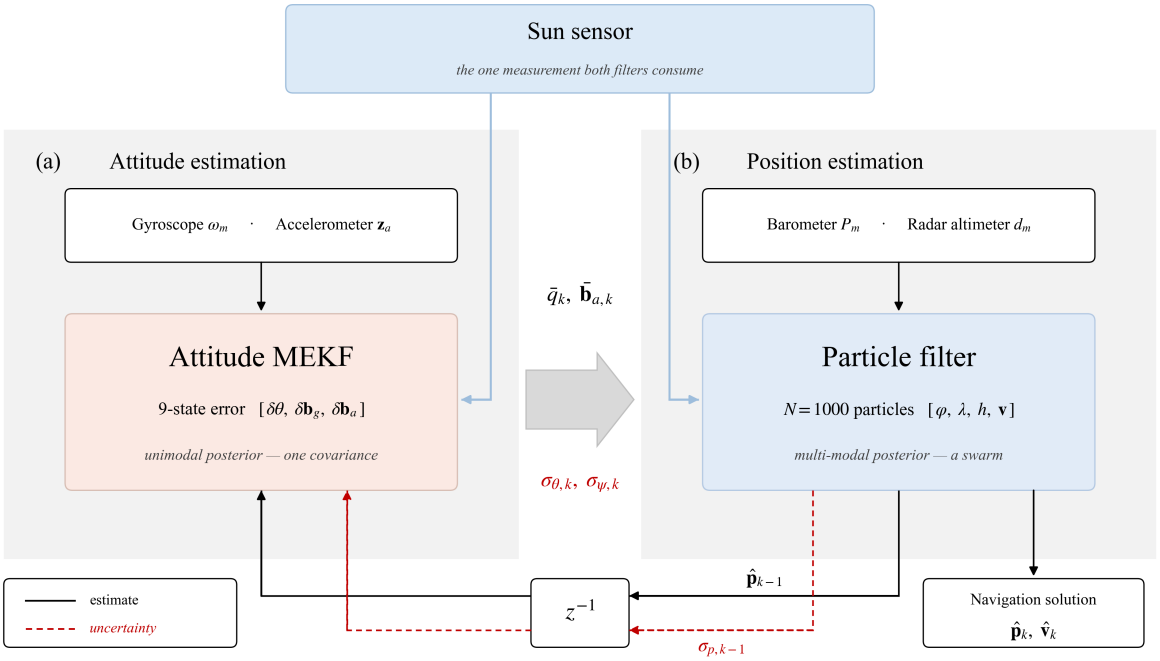


Figure 16. Coupled estimation architecture: attitude MEKF (a) and position particle filter (b). Solid arrows are estimates, dashed red arrows are uncertainties.

3.2 Attitude Estimation: the Multiplicative Extended Kalman Filter

3.3 Position Estimation: the Particle Filter

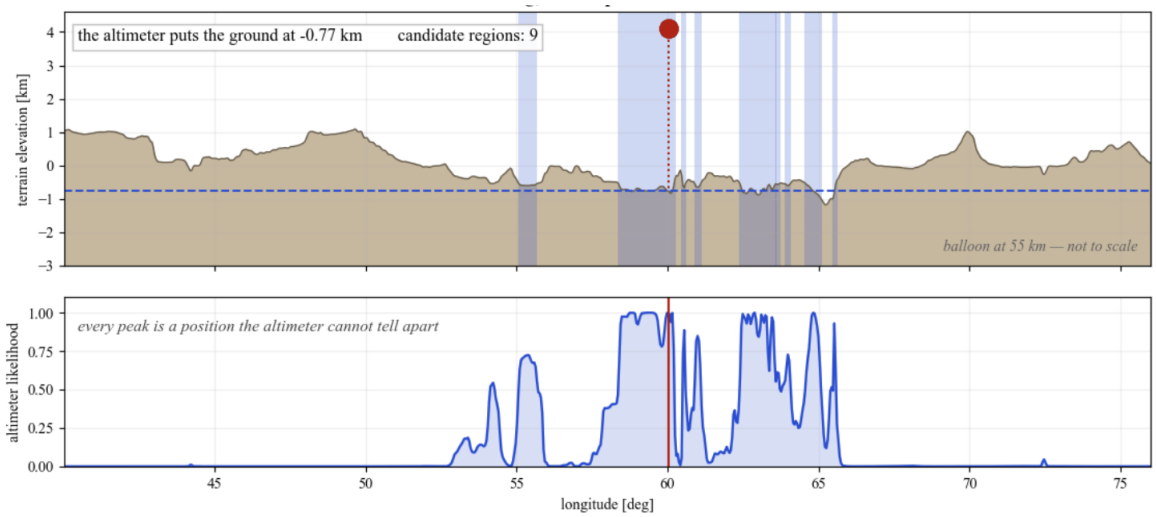


Figure 17. Multimodal posterior represented by the particle swarm: competing location hypotheses coexist as clusters until the measurements single one out.

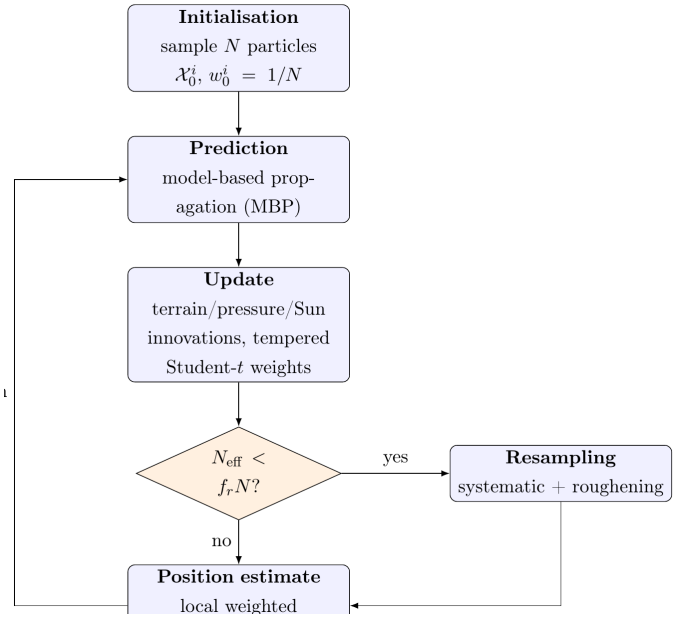


Figure 18. One cycle of the position particle filter: prediction, update with the robust Student-t likelihood, resampling with roughening when $N_{eff} < 0.5N$, position estimate.

3.4 Uncertainty Exchange Between the Two Filters

Table 2. Quantities exchanged between the two estimators at each epoch. Three of the five entries are uncertainties rather than values.

Quantity	Direction	Purpose in the receiving filter
$\bar{q}_k$	MEKF $\rightarrow$ PF	tilts the altimeter slant range; builds the measured horizon frame
$\bar{b}_{a,k}$	MEKF $\rightarrow$ PF	de-biases the specific force before the vertical is reconstructed
$\sigma_{\theta,k}$	MEKF $\rightarrow$ PF	inflates the Sun-channel noise
$\hat{p}_k$	PF $\rightarrow$ MEKF	builds the local-vertical and Sun references
$\sigma_{p,k}$	PF $\rightarrow$ MEKF	inflates the accelerometer and Sun measurement noise

3.5 The Longitude Problem and the Role of the Sun Sensor

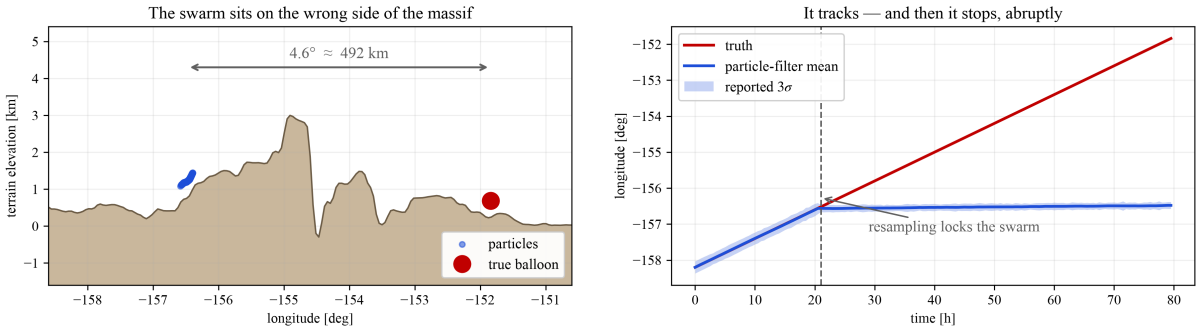


Figure 19. Terrain trapping with altimeter and barometer only: the estimate stays locked on one side of a massif (4 runs out of 10 diverge), motivating the Sun channel.

3.6 Results of a Single Reference Run

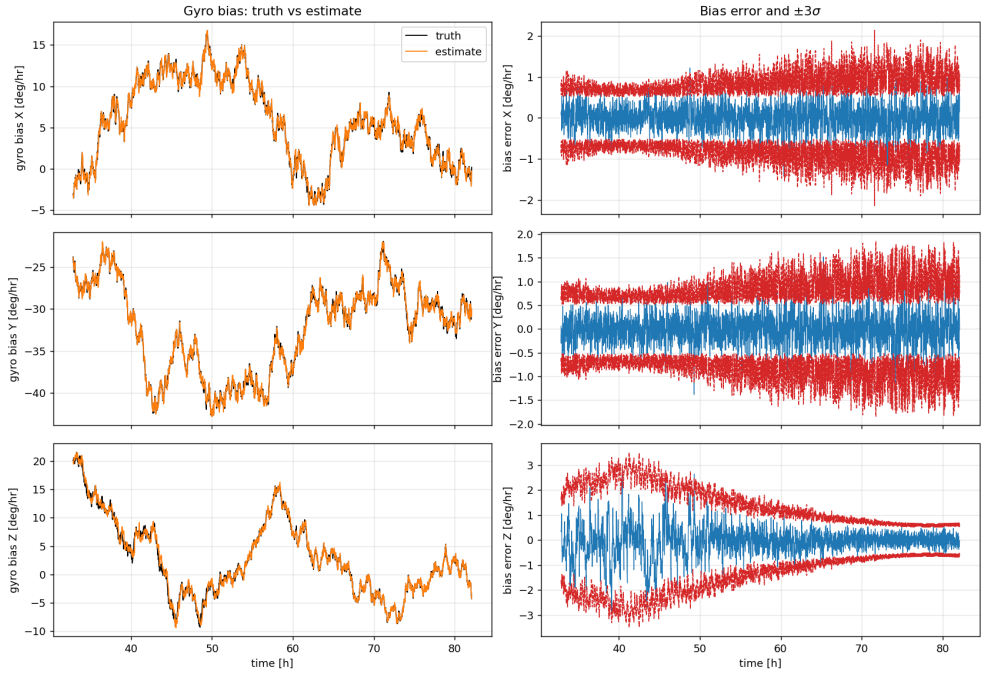


Figure 20. Single run: gyroscope bias, truth vs. MEKF estimate (left) and error against the declared $\pm 3\sigma$ (right).

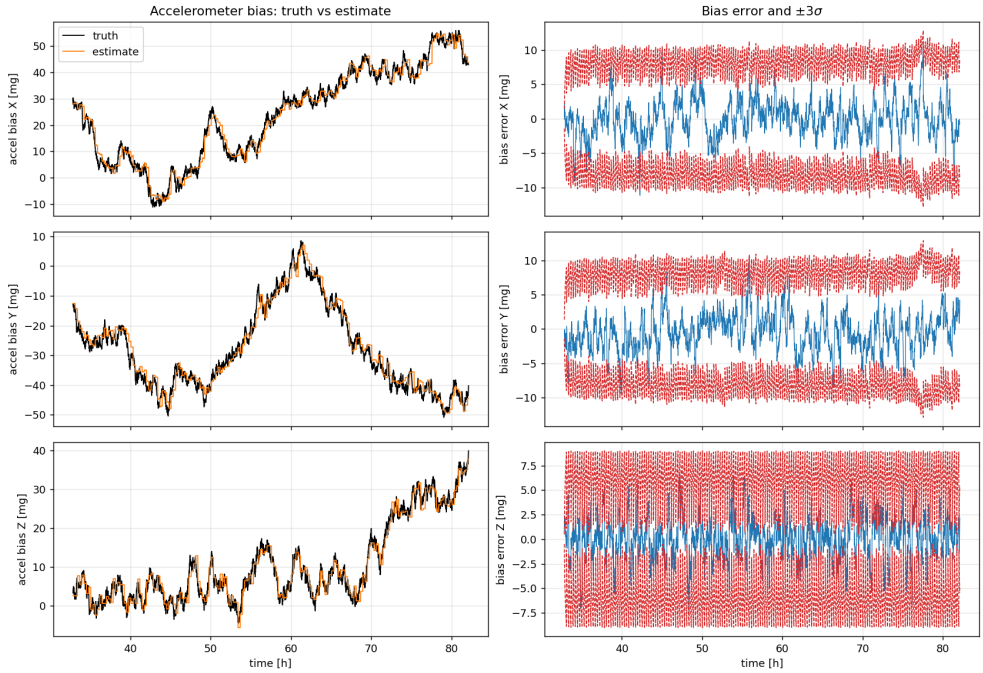


Figure 21. Single run: accelerometer bias, truth vs. MEKF estimate (left) and error against the declared $\pm 3\sigma$ (right).

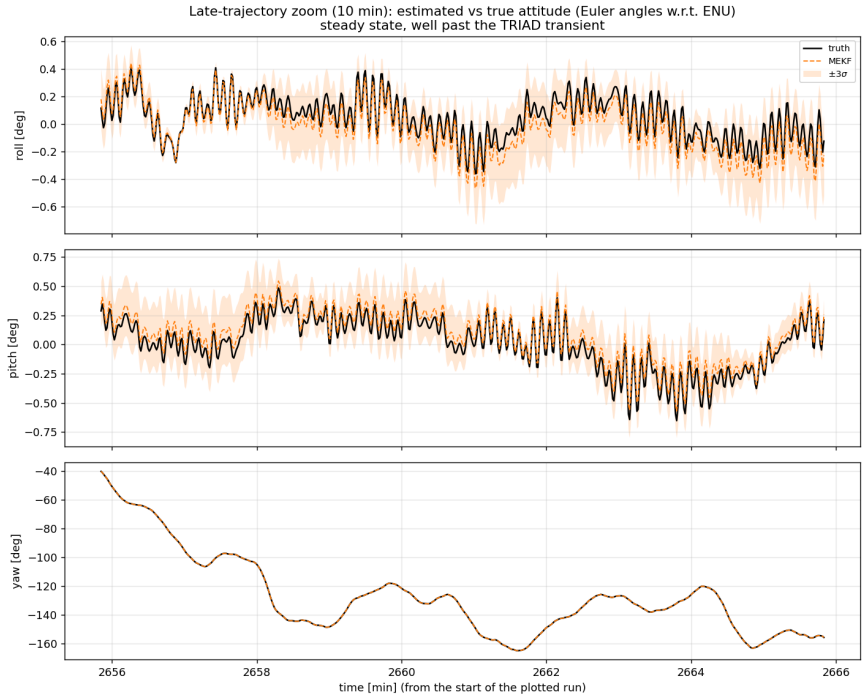


Figure 22. Single run: estimated vs. true roll, pitch and yaw over a 10 min window, with the MEKF $\pm 3\sigma$ band.

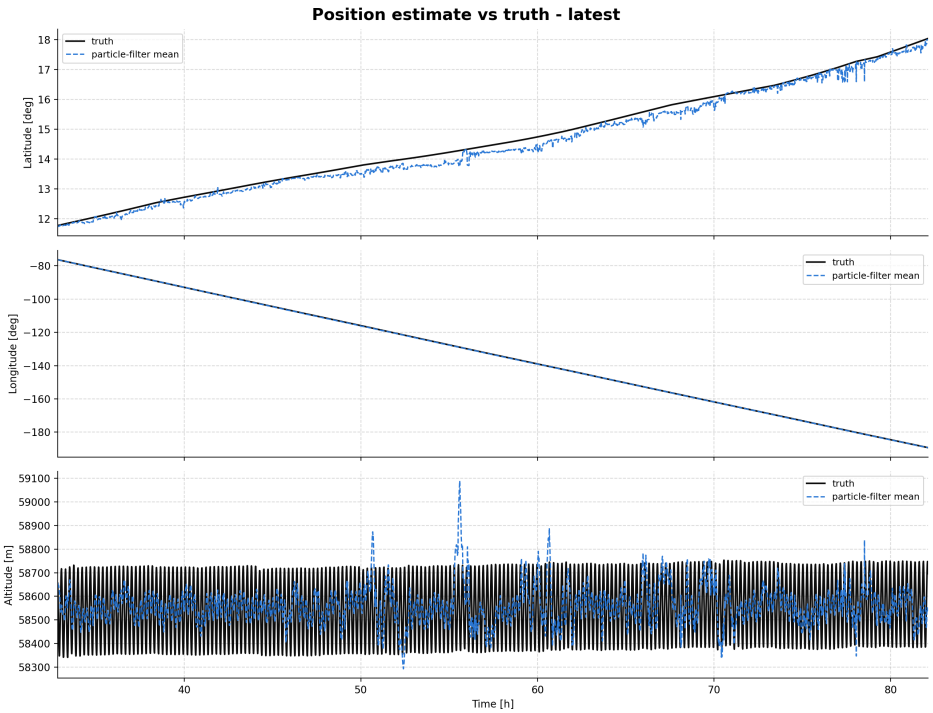


Figure 23. Single run: particle-filter position mean vs. truth (latitude, longitude, altitude).

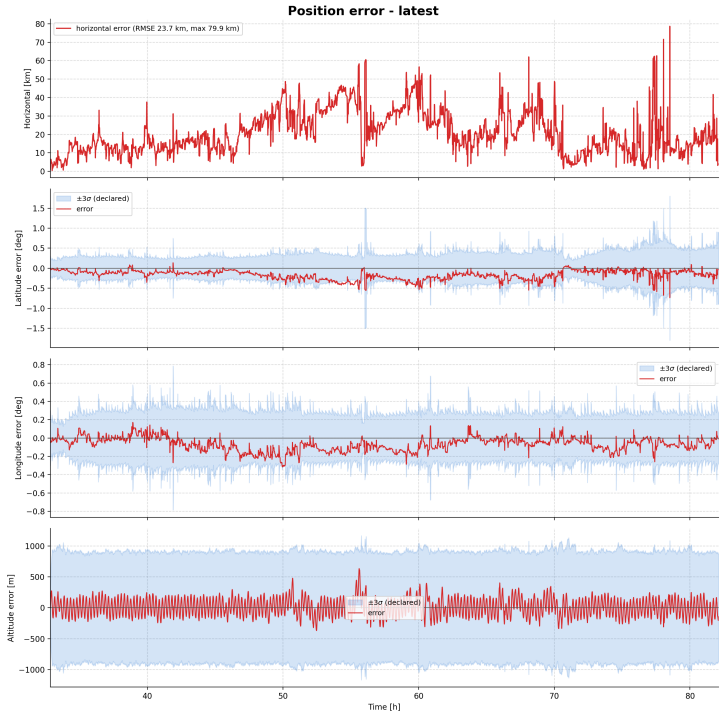


Figure 24. Single run: horizontal position error (RMSE 23.7 km, maximum about 80 km) and per-axis error against the declared $\pm 3\sigma$.

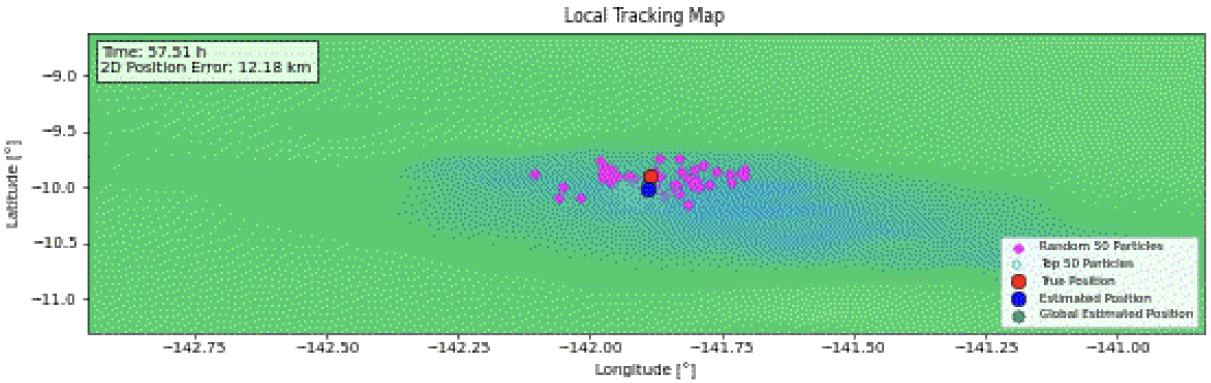


Figure 25. Particle swarm over the local topography at $t = 57.5$ h, with true and estimated positions.

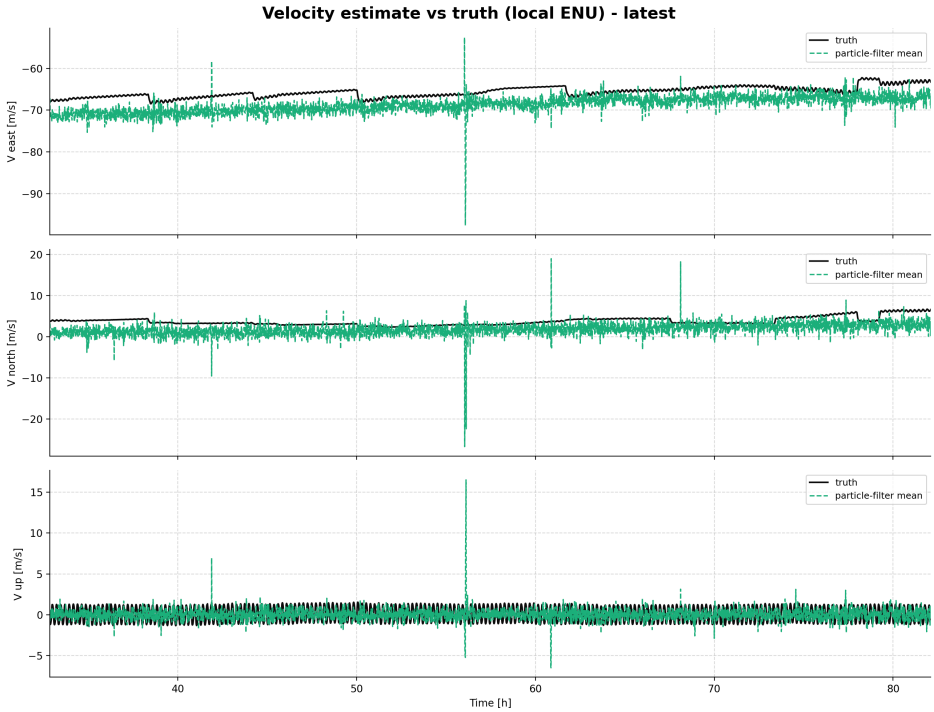


Figure 26. Single run: particle-filter velocity mean vs. truth in the local ENU frame.

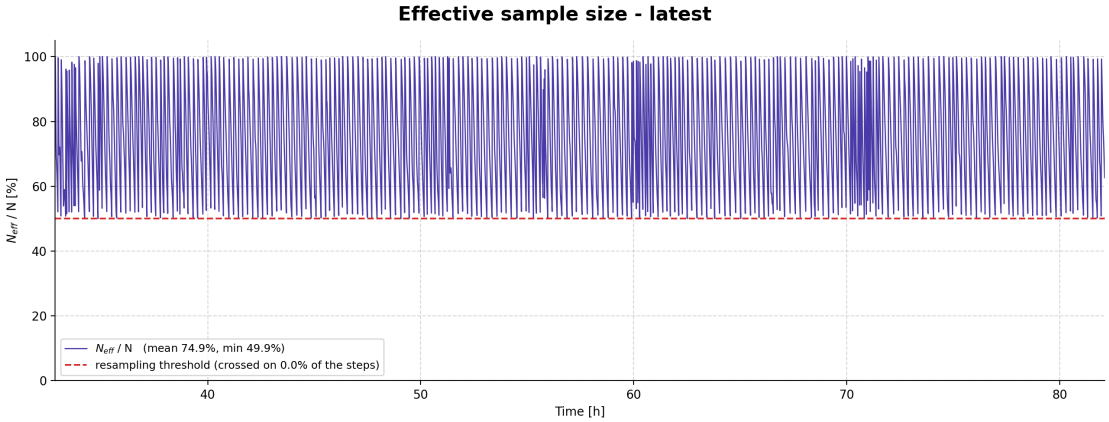


Figure 27. Effective sample size N_{eff}/N over the run, with the resampling threshold.

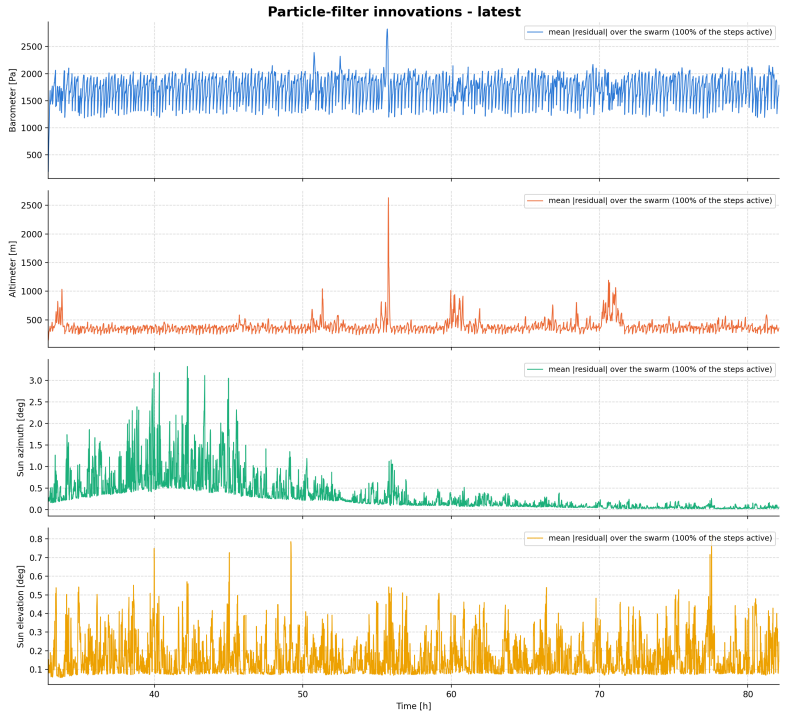


Figure 28. Mean absolute residual over the swarm for the four particle-filter channels (barometer, altimeter, Sun azimuth, Sun elevation).



Figure 29. Sensor usage over Sun-visible epochs: duty cycle, share of the particle filter's selection, and share over time.

4. SIMULATIONS AND RESULTS

4.1 Simulation Setup

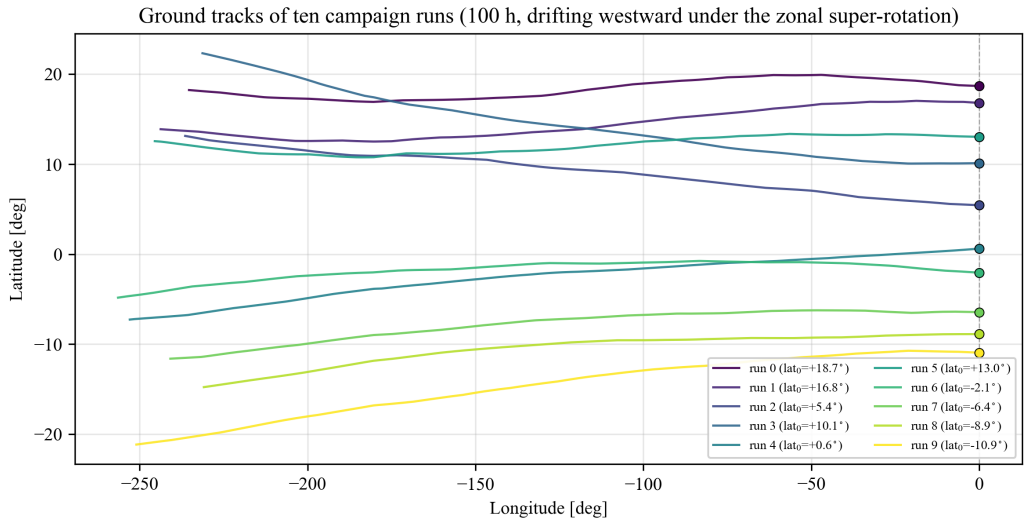


Figure 30. Ground tracks of 10 Monte Carlo runs, launched at longitude 0° with latitude uniform in $[-20°, 20°]$.

4.2 Sensitivity to the Atmospheric Model

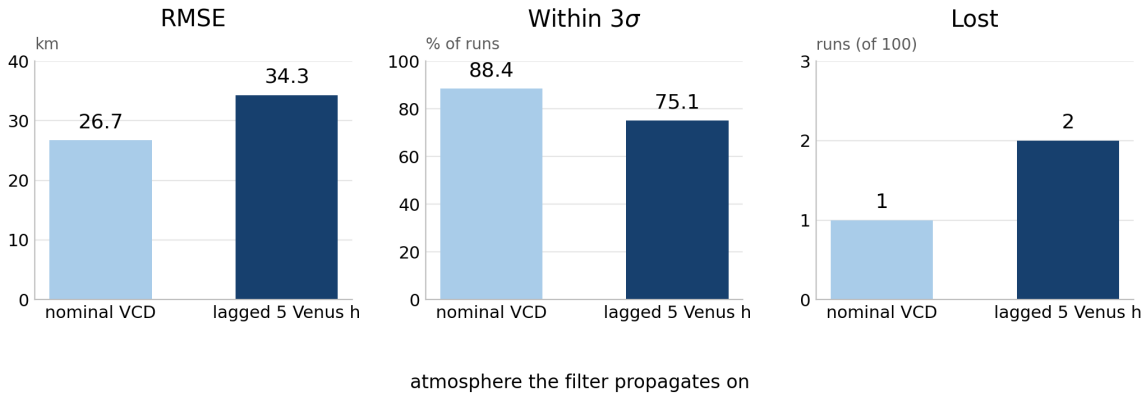


Figure 31. Atmosphere campaign (100 runs): reading the database five Venusian hours out of phase costs 7.6 km of RMSE (26.7 to 34.3 km).

Table 3. Sensitivity to the fidelity of the filter's onboard atmosphere. One hundred runs per row, sunlit arc only, identical seeds and initial latitudes.

Atmosphere the filter propagates on	RMSE [km]	3σ coverage [%]	Lost
Nominal database	26.7	88.4	1/100
Database lagged 5 Venus h	34.3	75.1	2/100

4.3 Sensitivity to the Sun Sensor Noise

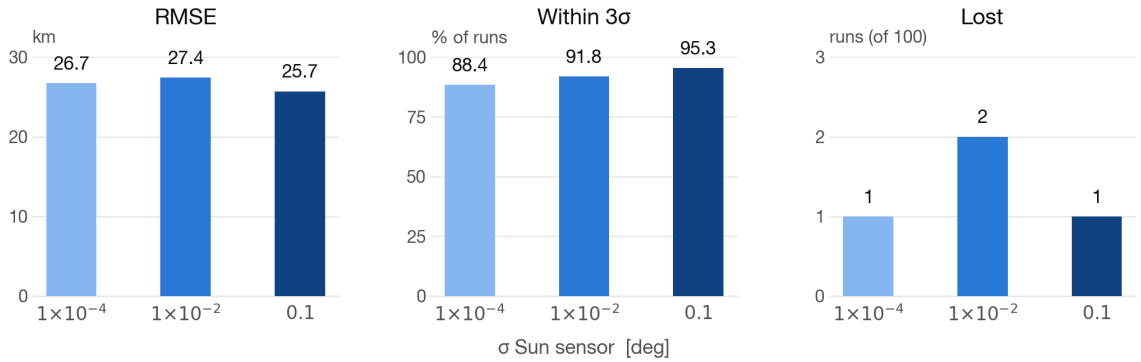


Figure 32. Sun-sensor campaign (100 runs per row): across three decades of noise the RMSE moves by only 1.7 km.

Table 4. Sensitivity to the Sun-sensor noise σ_s. One hundred runs per row, sunlit arc only, nominal atmosphere throughout.

σ_s	RMSE [km]	3σ coverage [%]	Lost
1×10^{-4} deg	26.7	88.4	1/100
1×10^{-2} deg	27.4	91.8	2/100
1×10^{-1} deg	25.7	95.3	1/100

5. CONCLUSIONS AND FUTURE WORK

5.1 Main Findings

5.2 Limitations and Future Work